Heart of Algebra

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

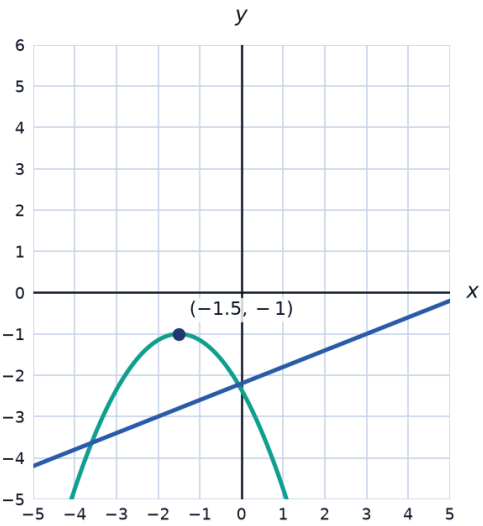
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Slope and equations of lines**

Find the slope of the line through $(2, 5)$ and $(6, 13)$.

2 **[Foundational] • Target: Slope and equations of lines**

Write the equation, in slope-intercept form, of the line with slope -3 passing through $(0, 7)$.



3 **[Foundational] • Target: Slope and equations of lines**

Find the slope of a line perpendicular to $y = \frac{2}{5}x - 1$.

4 **[Foundational] • Target: Systems of equations by substitution**

Solve the system: $y = 3x - 4$ and $2x + y = 11$.

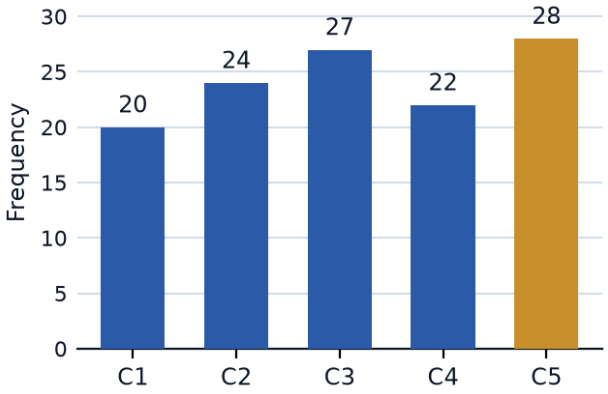
5 **[Foundational] • Target: Systems of equations by elimination**

Solve the system: $3x + 2y = 19$ and $x - 2y = 1$, using elimination.

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6 [Foundational] • Target: Interpreting linear models

A phone plan costs $C = 20 + 0.05m$ dollars for m minutes of calls. Interpret the meaning of the numbers 20 and 0.05 in this context.



7 [Foundational] • Target: Interpreting linear models

Company A charges $C = 30 + 0.20d$ and Company B charges $C = 0.35d$ for a d-mile trip. Find the mileage at which both companies charge the same amount, and state which company is cheaper for a 150-mile trip.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Systems with no solution or infinitely many solutions

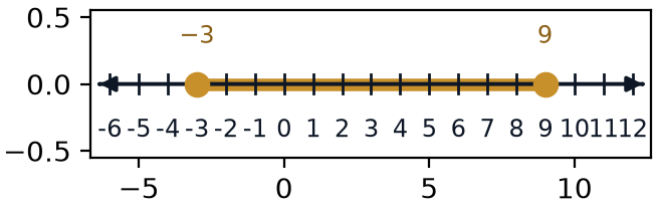
For what value of k does the equation $6x + 4 = kx + 4$ have infinitely many solutions?

9 [Intermediate] • Target: Systems with no solution or infinitely many solutions

Explain why the system $2x - 3y = 6$ and $4x - 6y = 10$ has no solution.

10 [Intermediate] • Target: Compound inequalities

Solve the compound inequality $-3 < 2x + 1 \leq 9$.



11 [Intermediate] • Target: Multi-step word problems

Two numbers have a sum of 50 and a difference of 12. Find the two numbers.

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12 [Intermediate] • Target: Multi-step word problems

The sum of three times a number and 7 equals 5 less than twice the number. Find the number.

13 [Intermediate] • Target: A well-designed word problem: ticket sales at a community theater

This question has three parts. A community theater sells adult tickets for 15 dollars and child tickets for 8 dollars. On Saturday the theater sold 120 tickets total and collected 1,450 dollars. (a) Set up a system of two equations for the number of adult tickets a and child tickets c sold. (b) Solve the system to find how many of each ticket type were sold. (c) On Sunday the theater raises the adult price to 18 dollars but keeps the child price the same. If the same number of adult and child tickets sell as on Saturday, find the new total revenue.

14 [Intermediate] • Target: Slope and equations of lines

For what value of the constant a will the system of equations $ax + 4y = 7$ and $3x + 2y = 5$ have no solution?

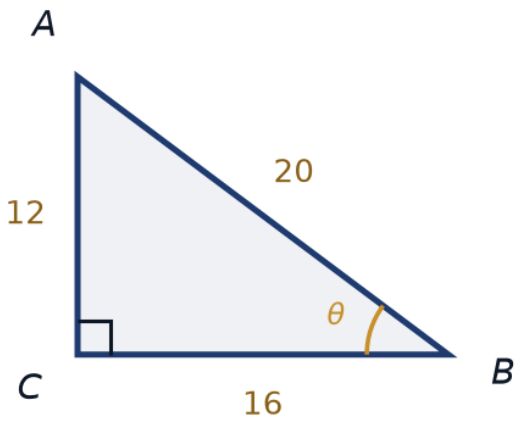
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Systems of equations by substitution • Cross-Lesson: 2.01 Quadratics & Polynomials

A school theater sold 250 tickets for a play, raising a total of $1,750$. $Adulttickets cost9$ each and student tickets cost 5 each. How many adult tickets were sold?

16 [Challenging] • Target: Systems of equations by elimination • Cross-Lesson: 3.01 Functions & Graphs

Level 4 · SynthesisIn the system of equations $ax + 8y = 24$ and $6x + by = 8$, a and b are constants. If the system has infinitely many solutions, what is the value of $a + b$?



17 [Challenging] • Target: Absolute value equations • Cross-Lesson: 4.01 Ratios, Rates & Data

Find all real solutions to the equation $|2x - 7| = 3x - 1$.

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18 [Challenging] • Target: Interpreting linear models • Cross-Lesson: 5.01 Statistics & Probability

Solve the linear system $3x + 2y = 16$ and $x - y = 2$ for x and y.

19 [Challenging] • Target: Systems with no solution or infinitely many solutions • Cross-Lesson: 6.01 Geometry

Level 2 · MediumA theater sold a total of 250 adult and student tickets for an evening performance, collecting $2,400$. $Ifadulttickets cost 12$ each and student tickets cost 8 each, how many student tickets were sold?

20 [Challenging] • Target: Compound inequalities • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Level 3 · AdvancedIn the hard Module 2, questions 18–22 are typically high-difficulty multi-step problems with parameters k. If a student has 7 minutes remaining for the last 3 questions, which approach optimizes score yield: attempting all 3 partially, or using Desmos regression and choice elimination to guarantee complete accuracy on 2 of them?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving linear equations

Two numbers have a sum of 63 and a difference of 9. Find the larger number.

a. 38

b. 36

c. 34

d. 27

22 [Foundational] • Target: Linear inequalities

For what value of constant k will the system of linear equations $3x - 5y = 8$ and $6x + ky = 16$ have infinitely many solutions?

a. -10

b. 10

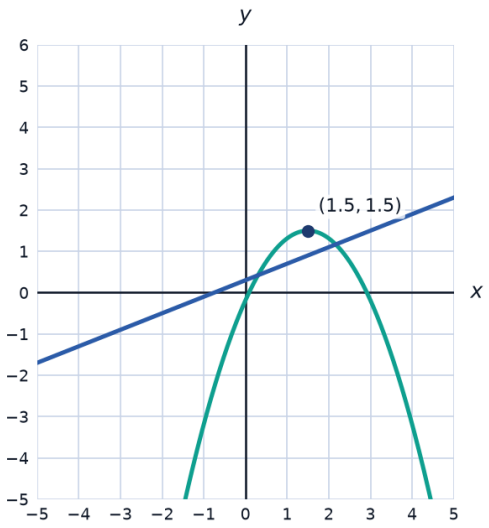
c. -5

d. 6

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23 [Foundational] • Target: Slope and equations of lines

In the system of equations below, c is a constant: $3x - 6y, = 9; x - 2y, = c$ If the system has infinitely many solutions, what is the value of c?



a. 3

b. 9

c. -3

d. 1

24 [Foundational] • Target: Systems of equations by substitution

A system of two linear equations has no solution. Which of the following must be true about the two lines?

a. They have the same slope but different y-intercepts

b. They have different slopes

c. They have the same slope and the same y-intercept

d. They are perpendicular

25 [Foundational] • Target: Systems of equations by elimination

In the system of equations below, k is a constant: $- - 4x + y, = 1; kx - 3y, = -15$ For which value of k does the system have no solution?

a. -4

b. 12

c. 3

d. -12

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26 [Foundational] • Target: Absolute value equations

A theater sells adult tickets for $x14andchildticketsfor\10$. On a night when 12 tickets were sold for a total of $x148$, how many adult tickets were sold?

a. 7

b. 5

c. 6

d. 8

27 [Foundational] • Target: Interpreting linear models

A theater sells large tickets for $x28andsmallticketsfor\9$. On a night when 16 tickets were sold for a total of $x201$, how many large tickets were sold?

a. 13

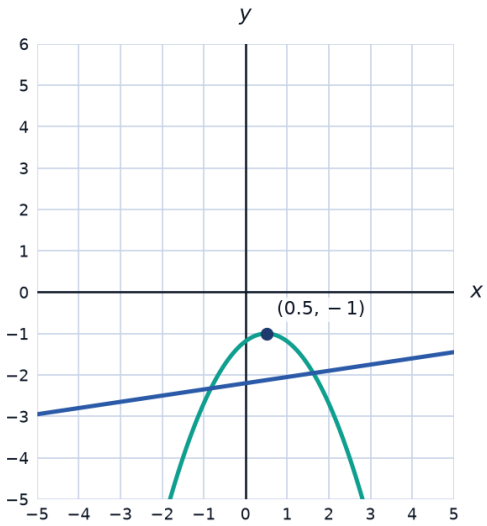
b. 3

c. 4

d. 2

28 [Foundational] • Target: Systems with no solution or infinitely many solutions

A theater sells premium tickets for $x8andstandardticketsfor\3$. On a night when 8 tickets were sold for a total of $x39$, how many premium tickets were sold?



a. 2

b. 5

c. 4

d. 3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Compound inequalities**

In the system $\{2x + py = 6; qx - 6y = 12$, there are infinitely many solutions. What is $p + q$?

a. -1

b. 0

c. 1

d. 2

30 **[Intermediate]** • **Target: Multi-step word problems**

For what value of the constant c does the system of equations $2x - 5y = 8$ and $6x - 15y = c$ have infinitely many solutions?

a. 16

b. 20

c. 24

d. 28

31 **[Intermediate]** • **Target: A well-designed word problem: ticket sales at a community theater**

In the system of equations: $\{4x + 6y = 15; 2x + ky = 7.5$ If the system has infinitely many solutions, what is the value of k?

a. 2

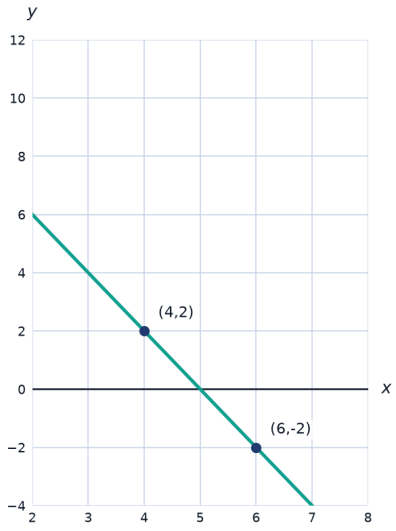
b. 3

c. 4

d. 6

32 **[Intermediate]** • **Target: Solving linear equations**

What is the slope of the line through $(4, 2)$ and $(6, -2)$?



a. -1/2

b. 2

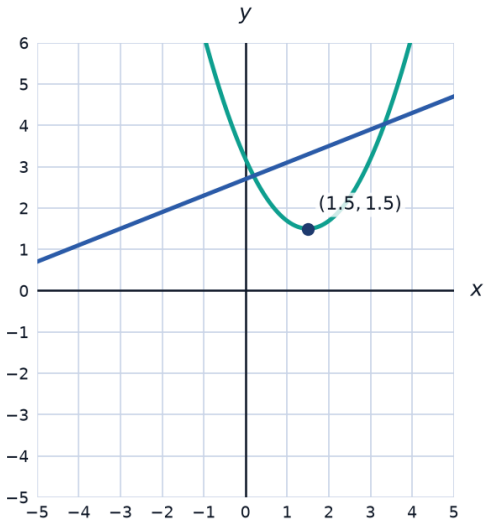
c. -1

d. -2

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33 **[Intermediate]** • **Target: Linear inequalities**

In the system $\{mx + 3y = 9; 4x + 6y = 18$, for what value of m does the system have infinitely many solutions?



a. 1

b. 2

c. 3

d. 4

34 **[Intermediate]** • **Target: Slope and equations of lines**

A particle's position in the xy-plane at time t is given by the parametric equations $x(t) = t^2 - 6t$ and $y(t) = t^3 - 9t$, where $t \geq 0$. The particle starts at the origin and moves along a path. Find the time $t > 0$ when the particle returns to the x-axis, and determine the x-coordinate of this position.

a. $t = 3$, x-coordinate = -9

b. $t = 4$, x-coordinate = -8

c. $t = 5$, x-coordinate = -5

d. $t = 6$, x-coordinate = 0

35 **[Intermediate]** • **Target: Systems of equations by substitution**

What is the sum of all solutions to the equation $|2x + 2| = 10$?

a. 2

b. 4

c. -6

d. -2

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36 [Intermediate] • Target: Systems of equations by elimination

Which ordered pair (x, y) satisfies the system of inequalities below? $y, > 3x + 1; y, \leq -x + 10$

a. $(0, -1)$

b. $(1, 6)$

c. $(6, 16)$

d. $(1, 1)$

37 [Intermediate] • Target: Absolute value equations

If $|3x - 5| = |x + 7|$, what is the product of all real solutions for x?

a. -3

b. 3

c. -6

d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Interpreting linear models • Cross-Lesson: 5.01 Statistics & Probability

What is the slope of the line given by the equation $4x - 2y = 10$?

a. 2

b. -2

c. 4

d. $\frac{1}{2}$

39 [Challenging] • Target: Systems with no solution or infinitely many solutions • Cross-Lesson: 6.01 Geometry

A line passes through the points $(2, 5)$ and $(6, 13)$. What is the slope of the line?

a. 2

b. 3

c. $\frac{1}{2}$

d. 4

40 [Challenging] • Target: Compound inequalities • Cross-Lesson: 7.01 Trigonometry & Additional Topics

What is the solution to the equation $2(x + 4) = 5(x - 2)$?

a. 6

b. 4

c. 8

d. 5

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41 [Challenging] • Target: Multi-step word problems • Cross-Lesson: 8.01 Exponentials & Logarithms

For what value of a does the system $ax + 4y = 7$ and $3x + 2y = 5$ have no solutions?

a. 6

b. -6

c. 3

d. 4

42 [Challenging] • Target: A well-designed word problem: ticket sales at a community theater • Cross-Lesson: 2.01 Quadratics & Polynomials

What is the sum of all solutions to $|5x - 7| = 10$?

a. 14/5

b. -3/5

c. 17/5

d. 24/5

43 [Challenging] • Target: Solving linear equations • Cross-Lesson: 3.01 Functions & Graphs

Solve the system of equations: $x + y = 14$ and $x - y = 6$. What is x?

a. 10

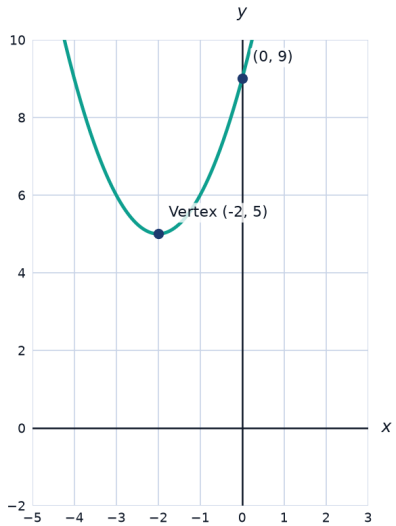
b. 4

c. 8

d. 12

44 [Challenging] • Target: Linear inequalities • Cross-Lesson: 4.01 Ratios, Rates & Data

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at the point $(-2, 5)$ and the parabola passes through the point $(0, 9)$. What is the value of b?



a. 4

b. 2

c. -4

d. -2

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45 [Challenging] • Target: Slope and equations of lines • Cross-Lesson: 5.01 Statistics & Probability

How many real solutions does the equation $|2x - 6| = x - 5$ have?

a. 0

b. 1

c. 2

d. Infinitely many

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